北京郵電大學

Queen Mary University of London

Joint Programme Assessments 2022/23

Answer Sheet

(To be used in conjunction with the provided alternative assessment questions and the supplementary instructions)

Complete the information below about yourself very carefully.

QM student number	200978947
BUPT student number	2020213248
Class number	2020215107
Module code	EBU5302
Word Count	690

I hereby confirm that the submitted work is entirely my own and I have not (i) used the services of any agency or person(s) providing specimen, model or ghostwritten work in the preparation of the work I submit for this open book assessment; (ii) given assistance in accessing this paper or in providing specimen, model or ghostwritten work to other candidates submitting for this open-book assessment.

Signature *Wu Tangxun*

For examiners ONLY Do not write in this section		Q1	Q2	Q3	Q4			Sum
	A							
	B							
	C							
	D							
	E							
	F							
	TOTAL							

Question	1. 2. 3.	Do not write in this column
Section(s)		

Question 1

The binary values: 110100101

Question 2

Data rate R = 3.2 Gb/s

l = 8 bits/level

therefore, D = R/l = 0.4 * (10^9) symbols/s

minimum B = 1/2*D = 0.2 GHz

Question 3

Firstly, we can see the equation of R(k) is:

$$R(k) = \sum_{i=1}^{L} (a_n a_{n+k}) P_i$$

And for bipolar RZ signalling the possible levels are +A, -A, 0 V.

Then, we can discuss as follows:

Assuming that the data are independent from bit to bit.

If k = 0, the possible products of a_n*a_n = 0, 0, A^2, (-A)^2

So R(0) = (1/2)*A^2

If k =1, the possible products of a_n*a_(n+k) = 0, 0, 0, -A^2

So R(1)= (-1/8)*A^2 + (-1/8)*A^2 = (-1/4)*A^2

If k>1, the possible products of a_n*a_(n+k) = 0, 0, 0, -A^2, A^2

So R(k)= (-1/16)*A^2 + (1/16)*A^2+ (-1/16)*A^2 +(1/16)*A^2 = 0

Overall,

$$R(k) = \begin{cases} \frac{A^2}{2}, & k=0 \\ -\frac{A^2}{4}, & |k|=1 \\ 0, & |k|>1 \end{cases}$$

Question	4. 5. 6. 7.	Do not write in this column
Section(s)		

Question 4

Bit rate R = 0.8 Mbits/s

Level L = 16

So l = log2(16) = 4 bits/level

Bandwidth B = 175 KHz

Because D = R/l = 200000 symbols/s

And D= 2B/(1+r), where r is the roll-off factor

Therefore 1+r = 2B/D = 1.75

And r = 0.75

Question 5

For thermal noise, Eb/N0=8.4dB

Temperature = 2900 K

Data rate = 2400bps

Because Eb/N0 = S dBW – 10lgR– 10lgK– 10lgT

So received signal level S = Eb/N0 + 10lgR + 10lgK + 10lgT

S = 8.4 + 10lg(2400) – 228.6 + 10lg(2900) = -151.774 dBW

Question 6

d = 1000m

f = 1 GHz

so $\lambda = c/f = 0.3m$

therefore, $L(dB) = 20lg(4 \pi d / \lambda) = 92.442$ dB

Question 7

There are 4 ways of combining signals for diversity in cellular system.

1. switch diversity: when one diversity of signal is too low, just try another.

2. selection diversity: when we want to combine signals, select and choose the largest signal for diversity.

3. Equal gain: all signals are equally weighted for combination and are added in phase.

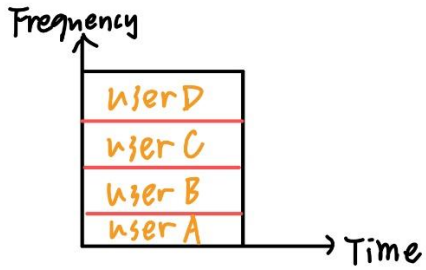
4. Maximal ratio: weigh the power in the branches in proportion to their signal amplitude and add in phase.

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FDMA: different frequency bands are assigned to different users.

Application scenarios: All 1G systems

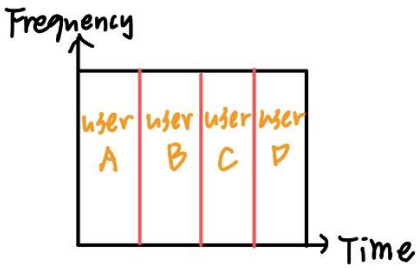
Diagrams:



TDMA: different time slots are assigned to different users.

Application scenarios: Most 2G systems

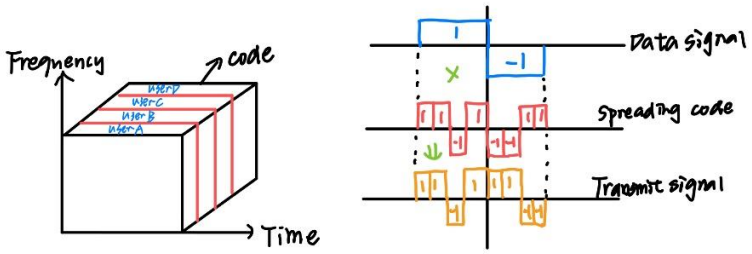
Diagrams:



CDMA: different codes are assigned to different users.

Application scenarios: some 2G systems and most 3G systems

Diagrams:



OFDMA: means orthogonal frequency division multiple access. It is a multi-user version of the popular OFDM digital modulation scheme. Multiple access is achieved in OFDMA by assigning subnets of subcarriers to individual users.

Application scenarios: 4G systems and 5G systems

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